

Universal Variable Kepler Solver

Assignment Presentation
Course: Celestial Mechanics

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- **Objective:** To implement a robust state propagator using the Universal Variable formulation to solve the Classical Two Body Problem.
- **Goal:** To achieve and maintain a numerical precision floor of 10^{-16} .
- **Scope:** Validating the solver across $0.1 \leq e \leq 2.5$ and $0.005 \leq h \leq 0.05$ using three numerical methods: Newton-Raphson, Bisection, Secant.

$$h = r_0 G_\alpha^1(s) + r_0 \dot{r}_0 G_\alpha^2(s) + k G_\alpha^3(s)$$

- h : timestep
- $r_0 = \|r_0\|$: initial radial distance
- $\dot{r}_0 = \frac{r_0 \cdot v_0}{r_0}$: radial velocity
- $k = GM$: gravitational parameter
- $\alpha = \frac{k}{a}$: energy parameter
- $G_\alpha^{(1)}(s)$, $G_\alpha^{(2)}(s)$, $G_\alpha^{(3)}(s)$: universal functions

G_n functions

- $G_0(s) = 1 - \alpha s^2 c(\alpha s^2)$
- $G_1(s) = s - \alpha s^3 s_{stumpff}(\alpha s^2)$
- $G_2(s) = s^2 c(\alpha s^2)$
- $G_3(s) = s^3 s_{stumpff}(\alpha s^2)$

The Stumpff Series

$$c_n(z) = \sum_{k=0}^{\infty} \frac{(-z)^k}{(n+2k)!}$$

- For Ellipses ($\alpha > 0$): They behave like trig functions (cos, sin).
- For Hyperbolas ($\alpha < 0$): They behave like hyperbolic functions (cosh, sinh).
- For Parabolas ($\alpha = 0$): They simplify to $c_0 = 1$, $c_1 = 1/2$, $c_2 = 1/6$, etc.
- Avoid the $0/0$ that occurs when the orbital energy α is zero.
- Code Implementation: For small z ($|z| < 10^{-6}$), the code uses a Taylor expansion to maintain precision, otherwise it uses trigonometric or hyperbolic closed forms depending on the sign of z .

- Universal Kepler Equation: $F(s) = r_0 G_1(s) + r_0 \dot{r}_0 G_2(s) + k G_3(s) - h = 0$
- Position update: $\mathbf{r}' = f \mathbf{r}_0 + g \mathbf{v}_0$
$$f = 1 - \frac{k}{r_0} G_2(s)$$
$$g = h - k G_3(s).$$
- Velocity update: $\mathbf{v}' = \dot{f} \mathbf{r}_0 + \dot{g} \mathbf{v}_0$
$$\dot{f} = -\frac{k}{r \cdot r_0} G_1(s)$$
$$\dot{g} = 1 - \frac{k}{r} G_2(s)$$

Validation

- Energy Conservation (E):

$$E = \frac{1}{2}v^2 - \frac{k}{r}$$

$$error = \frac{E - E'}{E}$$

- Angular Momentum ($\mathbf{l}$):

$$\mathbf{l} = \mathbf{r} \times \mathbf{v}$$

$$error = \frac{|\mathbf{l} - \mathbf{l}'|}{|\mathbf{l}|}$$

Newton-Raphson Iteration

- $s_{n+1} = s_n - \frac{F(s_n)}{F'(s_n)}$
- $F'(s) = r(s) = r_0 G_0^\alpha(s) + r_0 \dot{r}_0 G_1^\alpha(s) + k G_2^\alpha(s)$
- Advantage: Quadratic convergence.
- Disadvantage: Newton-Raphson requires an explicit formula for the derivative, so the derivation of may becomes more and more mathematically difficult.

Secant Method

- $s_{n+1} = s_n - F(s_n) \frac{s_n - s_{n-1}}{F(s_n) - F(s_{n-1})}$
- Advantage: Secant method approximates the slope using two previous points, so no derivative required.
- Disadvantage: susceptible to numerical round-off errors and can fail if the 2 estimated points become too close together near the root.

Bisection method

- Find $s_{mid} = \frac{s_{low} + s_{high}}{2}$
- Evaluate the sign of $F(s)$
- Advantage: Bisection method is absolutely robust, as long as the initial interval brackets the root, it cannot fail.
- Disadvantage: Slow.

Table: Baseline Simulation Parameters

Parameter	Symbol	Value	Unit
Central Mass	m_0	1.0	$M_\odot$
Orbiting Mass	m_1	0.001	$M_\odot$
Semi-major Axis	a	1.0	AU
Eccentricity	e	0.2	Dimensionless
Time Step	h (or Δt)	0.01	Years
Convergence Tolerance	τ	10^{-16}	Dimensionless

Results: Baseline Simulation

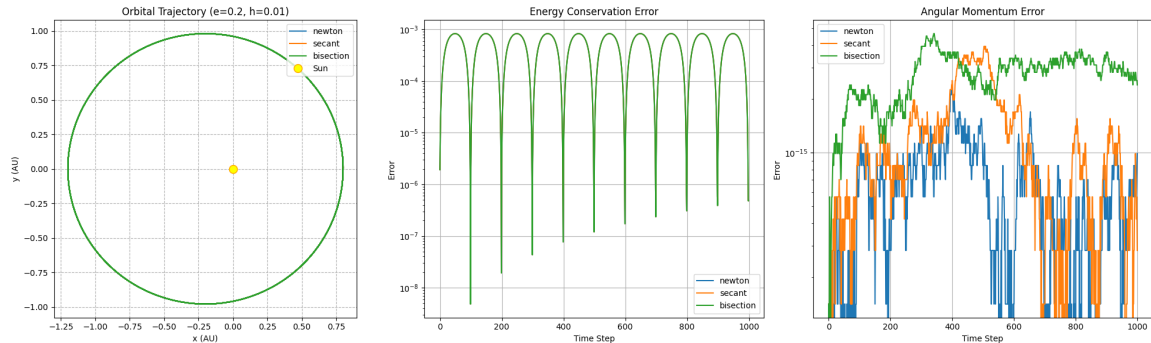


Figure: Simulation results for $e = 0.2$ and $h = 0.01$ comparing Newton-Raphson, Bisection, and Secant methods. The left panel shows the elliptical trajectory. The middle and right panels show the relative errors in energy and angular momentum respectively. The Sun is located at the origin $(0,0)$ in the (x, y) plane.

Results: Baseline Simulation

- All three methods (Newton, Secant, and Bisection) show the exact same orbit path.
- Energy error ranges between 10^{-3} and 10^{-8} .
- Small energy error spikes occur at periapsis.
- Angular momentum error ranges between 10^{-14} and 10^{-16} .

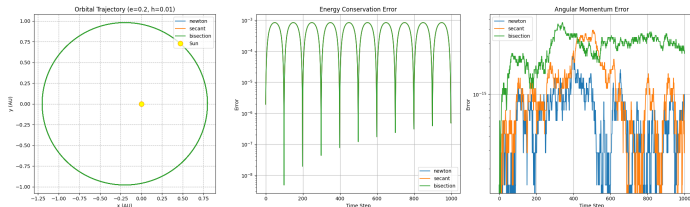


Figure: Simulation results for $e = 0.2$ and $h = 0.01$ comparing Newton-Raphson, Bisection, and Secant methods. The left panel shows the elliptical trajectory. The middle and right panels show the relative errors in energy and angular momentum respectively. The Sun is located at the origin (0,0) in the (x, y) plane.

Results: Mass Ratio Effect

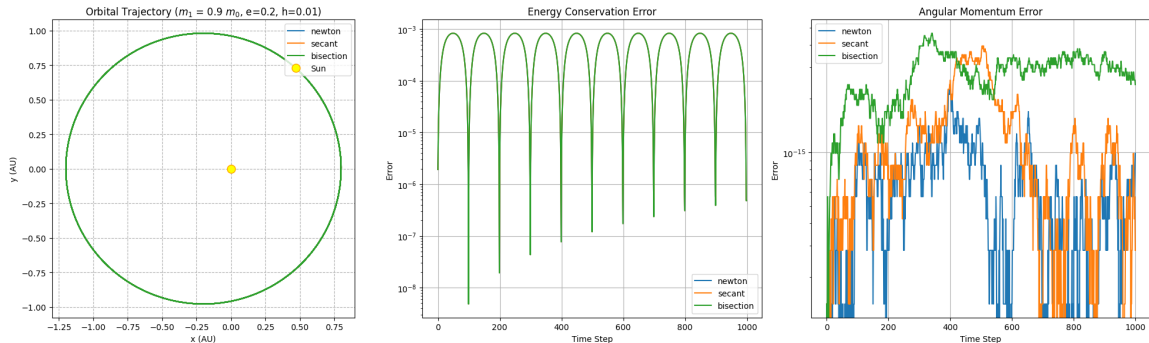


Figure: Simulation results for a high mass secondary ($m_1 = 0.9 M_{\odot}$).

No significant departure from the low mass baseline $\rightarrow$ mass does not affect stability of universal variable solver.

Results: Eccentricity Effect

- Elliptical ($e < 1.0$): Closed loops.
- Parabolic ($e = 1.0$): The critical boundary where an object has exactly escape velocity.
- Hyperbolic ($e > 1.0$): Open-ended trajectories (high energy flybys or interstellar objects).

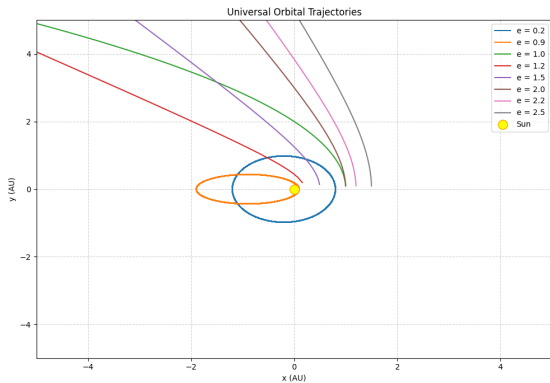


Figure: Comparison of orbital trajectories for eccentricities ranging from 0.2 to 2.5, fixed timestep $h = 0.01$ years. The Sun is located at the origin $(0, 0)$ in the (x, y) plane. (10000 iterations)

Results: Eccentricity Effect

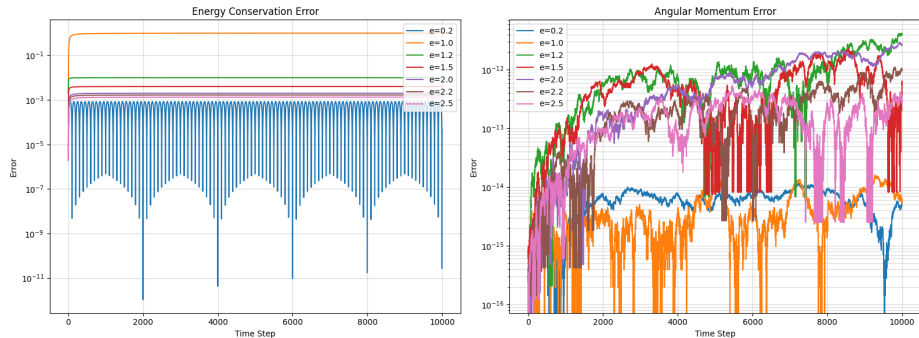


Figure: Energy and Angular Momentum Conservation Across $e = 0.2$ to 2.5.

$e \geq 1.0$ show a rapid initial error rise that later flattens $\rightarrow$ as $r \rightarrow \infty$, the gravitational gradient vanishes, and the solver transitions into a stable straight line propagation.

Results: Eccentricity Effect

As $e \rightarrow 1$, the compression of the ellipse along the y-axis follows $b = a\sqrt{1 - e^2} \rightarrow \text{flatten}$.

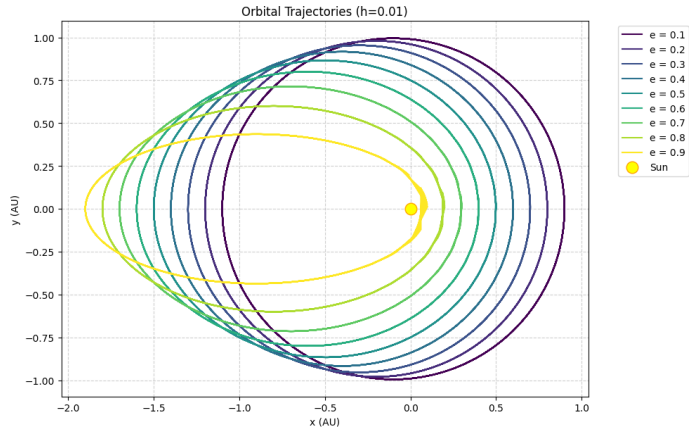


Figure: Comparison of orbital trajectories for eccentricities ranging from 0.1 to 0.9, fixed timestep $h = 0.01$ years. The Sun is located at the origin (0,0) in the (x, y) plane.

Results: Eccentricity Effect

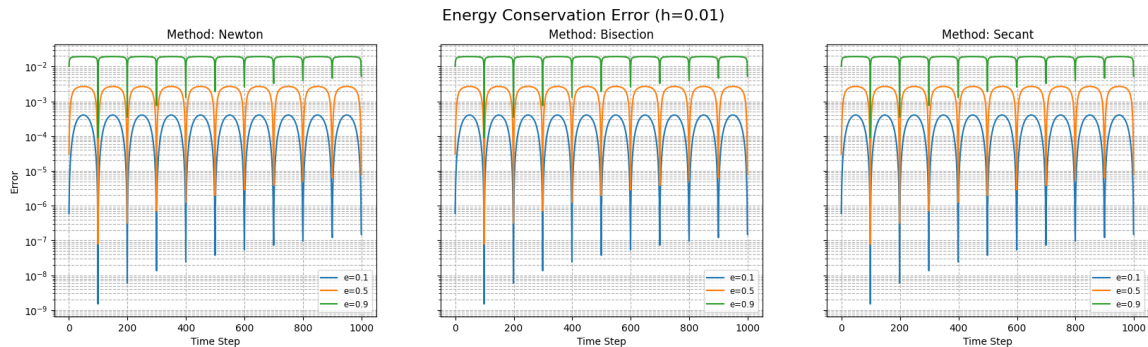


Figure: Relative energy conservation errors for fixed timestep $h = 0.01$ across eccentricities $e = 0.1, 0.5, 0.9$.

Results: Eccentricity Effect

- $10^{-2} \sim 10^{-9}$.
- Higher eccentric orbit $\rightarrow$ more error.
- Error reach $\sim 10^{-2}$ at periapsis (due to extreme acceleration and gravity at periapsis).

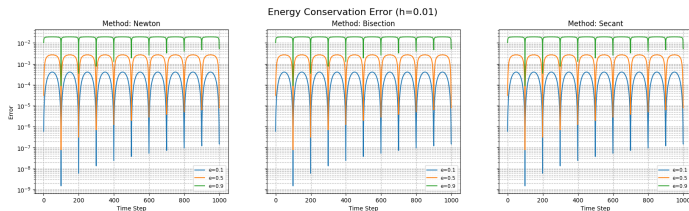


Figure: Relative energy conservation errors for fixed timestep $h = 0.01$ across eccentricities $e = 0.1, 0.5, 0.9$.

Results: Eccentricity Effect

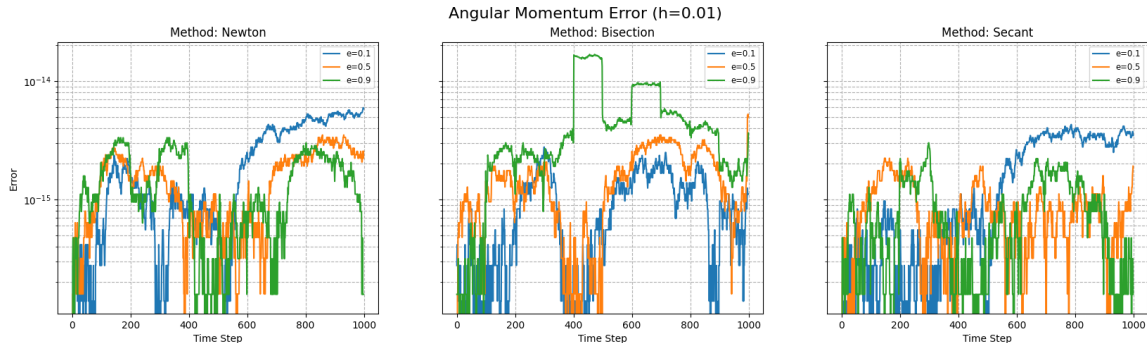


Figure: Relative angular momentum error for fixed timestep $h = 0.01$ across eccentricities $e = 0.1, 0.5, 0.9$.

Results: Eccentricity Effect

- $10^{-14} \sim 10^{-16}$.
- Higher eccentric orbit $\rightarrow$ more error.
- Bisection: highest error due to its slower convergence and rounding errors.

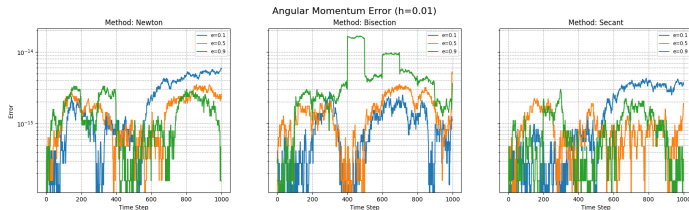


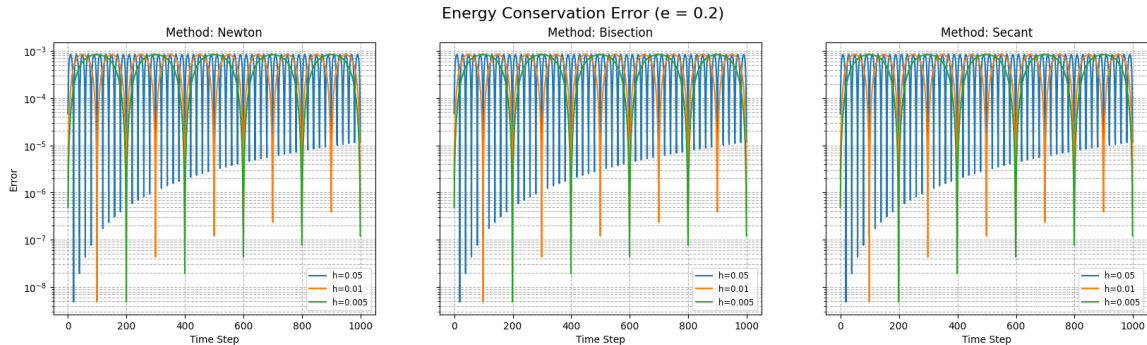
Figure: Relative angular momentum error for fixed timestep $h = 0.01$ across eccentricities $e = 0.1, 0.5, 0.9$.

Results: Eccentricity Effect

Table: Numerical methods performance over variable eccentricity (fixed $h = 0.01$)

e	Method	Max Energy Error	Max Momentum Error
0.2	Newton-Raphson	8.34585e-04	2.25314e-15
0.2	Secant	8.34585e-04	3.94299e-15
0.2	Bisection	8.34585e-04	4.64710e-15
0.5	Newton-Raphson	2.67469e-03	3.50506e-15
0.5	Secant	2.67469e-03	2.23049e-15
0.5	Bisection	2.67469e-03	5.25759e-15
0.9	Newton-Raphson	1.93143e-02	3.32365e-15
0.9	Secant	1.93143e-02	3.00712e-15
0.9	Bisection	1.93143e-02	1.67765e-14

Results: Timestep Effect



Results: Timestep Effect

- $10^{-3} \sim 10^{-8}$.
- All methods produce nearly identical error results.
- Smaller h , smaller error floor.
- The fluctuations are due to discretization of time into steps.

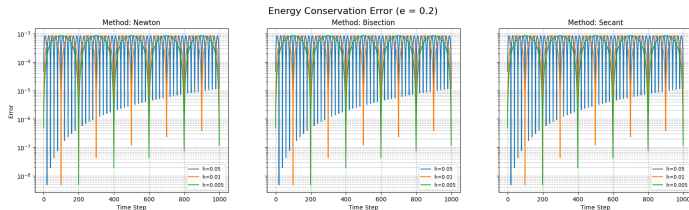


Figure: Comparative analysis of relative energy conservation error for $e = 0.2$ across three numerical methods (Newton, Bisection, Secant) and three temporal resolutions ($h = 0.05, 0.01, 0.005$).

Results: Timestep Effect

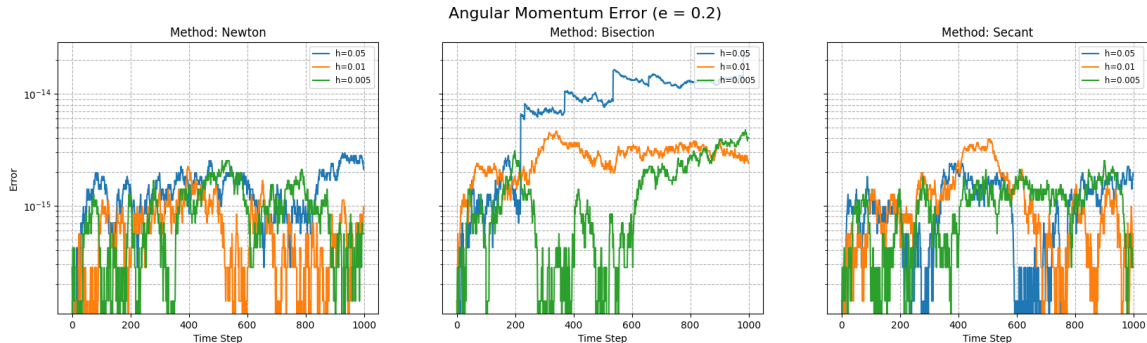


Figure: Comparative analysis of relative angular momentum error for $e = 0.2$ across three numerical methods (Newton, Bisection, Secant) and three temporal resolutions ($h = 0.05, 0.01, 0.005$).

Results: Timestep Effect

- $10^{-14} \sim 10^{-16}$
- Smaller h , smaller error floor.
- Bisection: highest error due to rounding.

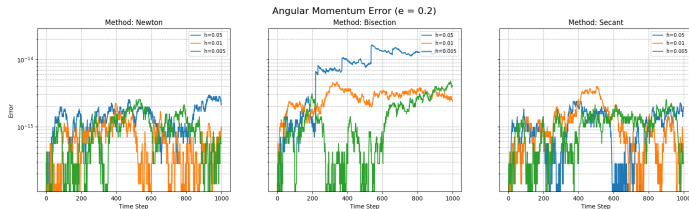


Figure: Comparative analysis of relative angular momentum error for $e = 0.2$ across three numerical methods (Newton, Bisection, Secant) and three temporal resolutions ($h = 0.05, 0.01, 0.005$).

Results: Timestep Effect

Table: Numerical methods performance over variable h (fixed $e = 0.2$)

h	Method	Max Energy Error	Max Momentum Error
0.05	Newton-Raphson	8.34585e-04	2.95724e-15
0.05	Secant	8.34585e-04	2.39396e-15
0.05	Bisection	8.34585e-04	2.25314e-14
0.01	Newton-Raphson	8.34585e-04	2.25314e-15
0.01	Secant	8.34585e-04	3.94299e-15
0.01	Bisection	8.34585e-04	4.64710e-15
0.005	Newton-Raphson	8.34585e-04	2.53478e-15
0.005	Secant	8.34585e-04	2.53478e-15
0.005	Bisection	8.34585e-04	4.78792e-15

Results

The best method: **Newton-Raphson**

- Efficiency: Converges to 10^{-16} in only ~ 5 iterations.
- Maintains the flattest error floor across different timestep and eccentricity regimes.
- Highest precision due to the use of the analytical derivative (physical radius r) to land on the root more cleanly with minimal rounding error.

NEWTON h	e	Max energy error	Max angular momentum error
0.05	0.2	8.34585e-04	2.95724e-15
0.05	0.5	2.67469e-03	3.98302e-15
0.05	0.9	4.18407e-02	2.79606e-02
0.01	0.2	8.34585e-04	2.25314e-15
0.01	0.5	2.67469e-03	3.50506e-15
0.01	0.9	1.93143e-02	3.32365e-15
0.005	0.2	8.34585e-04	2.53478e-15
0.005	0.5	2.67469e-03	3.02710e-15
0.005	0.9	1.93143e-02	3.79846e-15

SECANT h	e	Max energy error	Max angular momentum error
0.05	0.2	8.34585e-04	2.39396e-15
0.05	0.5	2.67469e-03	5.89487e-15
0.05	0.9	2.82077e-02	1.08022e-02
0.01	0.2	8.34585e-04	3.94299e-15
0.01	0.5	2.67469e-03	2.23049e-15
0.01	0.9	1.93143e-02	3.00712e-15
0.005	0.2	8.34585e-04	2.53478e-15
0.005	0.5	2.67469e-03	3.66438e-15
0.005	0.9	1.93143e-02	3.79846e-15

BISECTION h	e	Max energy error	Max angular momentum error
0.05	0.2	8.34585e-04	2.25314e-14
0.05	0.5	2.67469e-03	8.66387e-13
0.05	0.9	2.73400e-02	1.25937e-02
0.01	0.2	8.34585e-04	4.64710e-15
0.01	0.5	2.67469e-03	5.25759e-15
0.01	0.9	1.93143e-02	1.67765e-14
0.005	0.2	8.34585e-04	4.78792e-15
0.005	0.5	2.67469e-03	2.86778e-15
0.005	0.9	1.93143e-02	4.74808e-15

- Newton-Raphson is the most efficient numerical solver, reaching the 10^{-16} precision target in approximately 5 iterations.
- While Bisection is the most robust method, it is the least efficient approach due to its high number of iterations and rounded errors.
- Impact of Eccentricity e : Higher eccentricity ($e = 0.9$) leads to a higher energy conservation error floor compared to near-circular orbits ($e = 0.1$ and 0.2).
- Impact of Time Step h : Smaller values of h provide better energy conservation and less angular momentum error.

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- [4] K. Stumpff, Himmelsmechanik, vol. 1. Berlin, Germany: VEB Deutscher Verlag der Wissenschaften, 1962.
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Thank you for listening!